

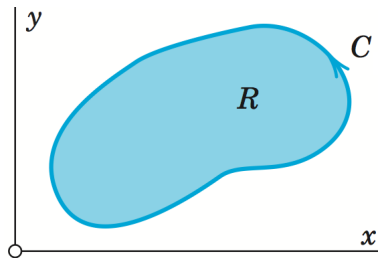
Topic: Green's Theorem

I. Introduction

In this section we will on line integrals over closed curves in the x, y -plane.

Notation: When representing a line integral over a closed curve, we will use the notation

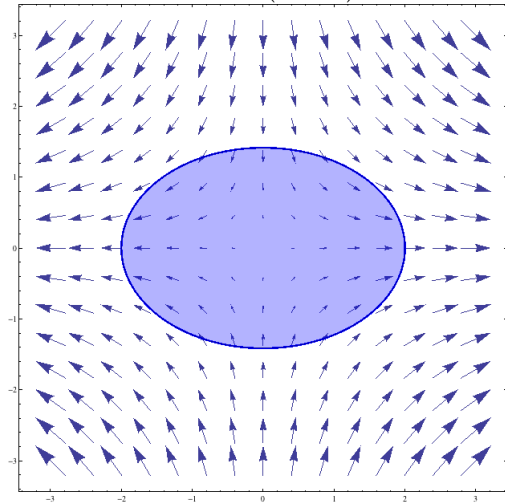
$$\text{instead of } \int_C \mathbf{F} \cdot d\mathbf{r}$$



Boundary of R is the Closed Curve C

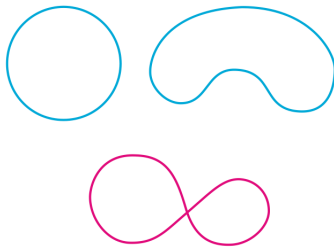
Note: Any planar closed curve can be thought of as the boundary of a region, R , in the plane.

F – Vector Field (Arrows)



For the rest of this section, the curves that we will consider have the property of being

Defn: A curve is said to be
if it never



Q: Is there a relationship between the region, R , the boundary, C , and vector field, F ?

A:

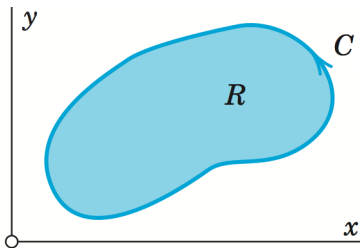
II. Green's Theorem

Thm: Let C is a simple closed curve bounding a planar region R . If $\mathbf{F}(x, y) = [M, N]$ is a vector field, then

$$\text{(General Form)} \quad \oint_C \mathbf{F} \cdot d\mathbf{r} =$$

Since $\mathbf{F} =$, a quick calculation shows: $(\text{curl } \mathbf{F}) \cdot \mathbf{k} =$, we have:

$$\text{(Component Form)} \quad \oint_C (M dx + N dy) =$$



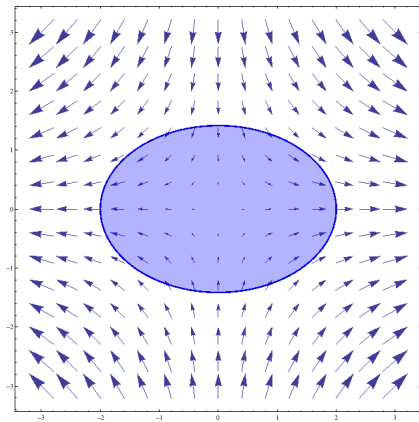
III. Forms of Green's Thm

Tangential Form:

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_R (\text{curl } \mathbf{F}) \cdot \mathbf{k} \, dA$$

$\mathbf{T}$ is the unit tangent vector along C .

Quantifies the tendency of the vector field to act in the direction of the boundary (circulation).

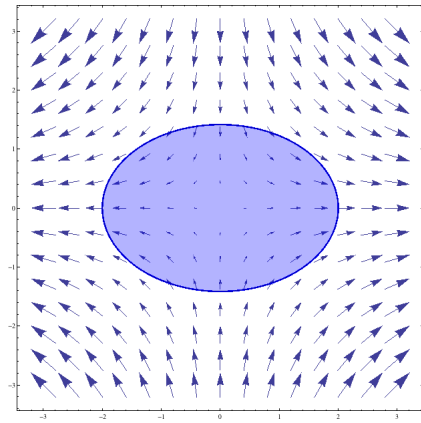


Normal Form:

$$\oint_C \mathbf{F} \cdot \mathbf{N} \, ds = \iint_R \text{div } \mathbf{F} \, dA$$

$\mathbf{N}$ is the unit normal vector along C .

Quantifies the tendency of the vector field to act perpendicularly to the boundary (flux).



Before verifying these let's summarize some notation:

(i) $\mathbf{r}'(t)$

(ii) $\mathbf{T}$

(iii) $\mathbf{N}$

(iv) $d\mathbf{r}$

(v) ds

Tangential Form:
$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_R (\text{curl } \mathbf{F}) \cdot \mathbf{k} \, dA$$

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds =$$

Normal Form:
$$\oint_C \mathbf{F} \cdot \mathbf{N} \, ds = \iint_R \text{div } \mathbf{F} \, dA$$

$$(*) \quad \mathbf{N} \, ds = (\mathbf{T} \times \mathbf{k}) \, ds$$

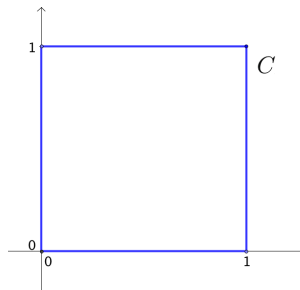
and so, by (*), we have ...

Ex: Let $\mathbf{F} = [x^2 + 4y, x + y^2]$ and let C be a square bounded by: $x = 0$, $x = 1$, $y = 0$, and $y = 1$,

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

To compute this directly we would have to compute

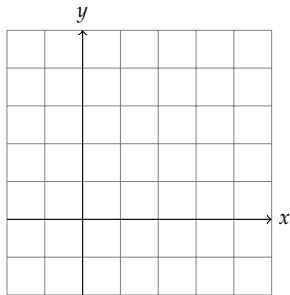
$$\oint_C \mathbf{F} \cdot d\mathbf{r} =$$



Ex: Let $\mathbf{F} = [16xy + 3yx^2, x^3 + 2y]$ and let C be a triangle bounded by: $x = 3$, $y = 0$, and $y = x$

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

First compute:



Ex: Let $\mathbf{F} = [xy, y^2]$ and let R be the region in the first quadrant bounded by: $y = x^2$, and $y = x$

Compute $\oint_C \mathbf{F} \cdot \mathbf{T} \, ds$ and $\oint_C \mathbf{F} \cdot \mathbf{N} \, ds$

Ex: Let $\mathbf{F} = \nabla(x^2 \sin y)$ and let R be the region bounded by: $x = 1$, $y = 0$, and $y = \sqrt{x}$

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

Summary: Line Integral over Vector-Valued Functions

We have 3 main methods for computing line integrals over a vector-valued function $\mathbf{F}$:

- (1) Directly
- (2) Fundamental Thm of Line Integrals
- (3) Green's Thm

C closed?	$\mathbf{F}$ Conservative? ($\text{curl } \mathbf{F} = \mathbf{0}$?)	Use Method	$\mathbf{r}$ needed?
Yes	Yes		No
	No		No
No	Yes		No
	No		Yes

Note: Method (1) can always be applied to any line integral